

15. A point source of sound emits energy equally in all directions at a constant rate and a detector placed 3.0 m from the source measures an intensity of 3.0 W m^{-2} .

What intensity would the detector measure if it is now placed at a distance 5.0 m from the source?

A 1.1 W m^{-2} **B** 1.8 W m^{-2} **C** 2.5 W m^{-2} **D** 8.3 W m^{-2}

Ans: A

For a point source of sound that emits energy equally in all directions,

Intensity at a point, $I \propto \frac{1}{r^2}$ where r is the distance between the point and the source.

$$\text{Therefore, } \frac{I'}{I} = \left(\frac{r}{r'} \right)^2$$

$$\Rightarrow \frac{I'}{3.0} = \left(\frac{3.0}{5.0} \right)^2$$

$$\therefore I' = \left(\frac{3.0}{5.0} \right)^2 \times 3.0 = 1.1 \text{ W m}^{-2}$$

16 A musical organ produces notes by blowing air into a set of pipes that are open at